Document 25: Green Campus, Sustainability & Hazardous Waste Management

Institution: Hyderabad Institute of Technology (HIT)

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1. Statutory Environmental Mandate and Institutional Goals

The Hyderabad Institute of Technology (HIT) is committed to operating an environmentally sustainable, net-zero-energy academic facility. In compliance with the guidelines established by the Central Pollution Control Board (CPCB) and the Telangana State Pollution Control Board (TSPCB), the university enforces strict ecological and waste-management regulations across all academic blocks, research centers, dining facilities, and student residential halls.

The Department of Campus Sustainability and Environmental Safety (DCSES) is the regulatory authority responsible for auditing institutional energy consumption, maintaining the campus biological reserve, and managing all municipal, electronic, and hazardous waste streams.

Every enrolled student, faculty member, and campus contractor is legally bound by these sustainability regulations. The university operates on the statutory principle that environmental contamination and illegal waste dumping represent direct violations of the institutional charter, subject to swift punitive action and state regulatory reporting.

2. Single-Use Plastic Ban and Sustainable Dining Directives

To reduce non-biodegradable landfill accumulation, the university enforces a campus-wide ban on single-use plastics.

The procurement, possession, distribution, and commercial sale of single-use plastic items are banned within the campus perimeter. Prohibited items include:

- Plastic water bottles with a volume under 1 liter.
- Plastic carry bags of any thickness.
- Single-use plastic straws, stirrers, plates, cups, and cutlery.
- Plastic cling-film and polystyrene (thermocol) packaging.

Enforcement in Canteens and Student Events:

All commercial campus canteens, third-party food stalls, and student festival organizers must utilize biodegradable dining alternatives, such as areca palm plates, pressed sugarcane bagasse containers, and stainless-steel cutlery.

If a student club or vendor is caught distributing single-use plastic water bottles or food containers during a workshop or cultural event:

- **First Infraction:** A spot environmental penalty of ₹2,500 is levied against the organizing entity, and all prohibited plastics are confiscated for recycling.
- **Second Infraction:** The fine escalates to ₹10,000, and the student club's room booking privileges are revoked for one full semester.

Students are expected to bring personal, reusable steel or glass water bottles, which can be refilled at the purified reverse-osmosis (RO) water stations installed across all building corridors.

3. Solid Waste Segregation Architecture and Color-Coding Matrix

The university operates a four-stream decentralized waste segregation model to achieve zero-landfill diversion. Standard mixed trash bins have been removed across the entire campus; all waste must be separated at the point of disposal into dedicated, color-coded collection receptacles.

Table 3.1: Color-Coded Solid Waste Segregation Protocols

Bin Color	Waste Classification	Permitted Waste Items	Prohibited Waste Items
Green Bin	Organic / Wet Waste	Food scraps, fruit peels, tea bags, leftover mess food, dry garden leaves	Plastic wrappers, metal cans, glass, tissues
Blue Bin	Dry Recyclables	Paper, cardboard notebooks, clean aluminum cans, intact glass bottles	Soiled food containers, wet organic waste, batteries
Red Bin	Domestic Hazardous	Broken glass shards, aerosol spray cans, sanitizing chemical wipes, fluorescent tube shards	Standard paper, unsoiled recyclables, organic matter
Yellow Bin	Sanitary / Biomedical	Used bandages, sanitary napkins, adhesive	General plastics, scrap metal,

		medical plasters, disposable masks	hardware components

Disposing of wet food waste into a Blue Recyclable bin or tossing broken glass into a Green Organic bin compromises the entire waste processing batch. Campus housekeeping teams log repeat violations in specific hostel corridors. If a hostel wing repeatedly contaminates segregation bins, the entire wing is charged a shared collective cleaning levy of ₹1,500 per incident.

4. Laboratory Chemical and Biological Waste Disposal

Academic research and student laboratory practicals produce significant volumes of spent chemical solvents, heavy metals, and microbiological cultures. Improper disposal of laboratory waste into the municipal sewage grid poses severe ecological and legal hazards.

Under no circumstances may students pour spent chemical reagents, acidic solutions, organic solvents, or chemical stains down the standard sink drains. Every wet laboratory contains specialized, labeled, heavy-duty polypropylene carboys for chemical waste collection:

- **Carboy A (Halogenated Solvents):** Dichloromethane, chloroform, and carbon tetrachloride residues.
- **Carboy B (Non-Halogenated Solvents):** Acetone, methanol, ethanol, and ethyl acetate residues.
- **Carboy C (Aqueous Heavy Metals):** Lead, mercury, chromium, and copper solutions.
- **Sharps Disposal Containers:** Syringes, scalpels, glass Pasteur pipettes, and broken microscope slides must be placed exclusively in puncture-proof yellow sharps containers.

Microbiological Waste Neutralization:

Before disposal, all biological petri dishes, agar slants, and bacterial culture media must undergo steam autoclaving at 121°C (250°F) at 15 psi for a minimum of 30 minutes to ensure complete pathogen neutralization. Students caught discarding un-autoclaved biological cultures or dumping solvents into sinks will receive an immediate F-grade for the laboratory course and be permanently barred from conducting research in departmental laboratories.

5. Electronic Waste (E-Waste) Management and Project Scrapping

Engineering projects, hardware prototyping, and IT infrastructure upgrades generate substantial quantities of Electronic Waste (E-Waste). The disposal of electronic components with municipal solid waste is illegal under national E-Waste Management Rules.

Prohibited E-Waste Items in Standard Trash:

- Printed Circuit Boards (PCBs) and breadboards.
- Lithium-ion, nickel-cadmium, and lead-acid batteries.
- Microcontrollers (Arduino, Raspberry Pi, ESP modules).
- Soldering wire fragments, copper wiring, and damaged sensors.
- Cathode-ray tubes, LED monitors, and computer power supply units (PSUs).

The Departmental E-Waste Deposit Protocol:

At the conclusion of the eighth-semester Major Final Year Project, teams dismantling hardware prototypes must separate operational components from unusable scrap. Operational parts can be donated to the Junior Component Library. Unusable electronics must be surrendered to the Departmental E-Waste Vault located in the Engineering Workshop block.

The E-Waste Vault issues a digital "E-Waste Disposal Memo" to the project team. This memo is a mandatory document required by the Departmental Project Review Committee (PRC) to clear the final project sign-off. If a project team discards electronic scrap into standard campus dustbins or abandons a hardware frame in a hallway, the team will be fined ₹5,000, and their project viva results will be frozen.

6. Energy Conservation Mandates and Hostel Power Quotas

To maintain its green building accreditation, HIT implements automated energy conservation protocols across all academic and residential spaces.

- **Hostel Energy Quotas:** Rooms are allocated a baseline electricity consumption allowance of 120 kilowatt-hours (kWh) per month for double rooms and 180 kWh per month for triple rooms. Sub-meters installed outside each room track consumption in real-time. If a room exceeds its monthly energy threshold due to leaving lights, fans, or authorized grooming appliances running continuously, the excess consumption is billed directly to the occupants at a commercial tariff rate of ₹12 per unit (kWh).
- **Academic Block Smart Controls:** Academic classrooms and lecture halls operate on centralized infrared occupancy sensors. If a room detects zero motion for more than 10 continuous minutes, the smart automation grid cuts power to all lights, fans, and projection units. Students are prohibited from bypassing or tampering with room sensors to keep empty classrooms energized.
- **HVAC Thermostat Lock:** In centralized air-conditioned spaces, including computer laboratories, the central library, and administrative offices, thermostats are locked at a mandatory cooling setpoint of 24°C (75.2°F). Faculty and students cannot override these controls to lower the temperature, ensuring compliance with national energy conservation efficiency norms.

7. Flora and Fauna Protection and Campus Ecological Reserves

The university campus encompasses a dedicated 15-acre protected ecological green zone that serves as a wildlife habitat and groundwater recharge catchment area.

Students are strictly prohibited from:

- Damaging, carving graffiti into, or cutting branches from trees on the campus grounds.
- Disturbing, capturing, feeding, or harassing native campus wildlife, including peacocks, spotted deer, monitor lizards, and resident bird species.
- Littering within the designated botanical garden and forest jogging trails.

Any student caught harming campus flora or fauna, driving two-wheelers through off-road ecological conservation zones, or dumping chemical materials in natural waterways will face the Disciplinary Action Committee (DAC). The DAC is empowered to levy environmental restoration fines up to ₹25,000 and refer the violation to the State Forest and Wildlife Department for legal prosecution under the Wildlife Protection Act.